

Probabilistic Forecasting of Sea Level Anomaly and Significant Wave Height Over the Indian Ocean

Bhavya Boda

202301037

B.Tech ICT

Tirth Gandhi

202301016

B.Tech ICT

Course: PC406 — BTech Mini Project 1

Faculty Mentor: Prof. Pritam Anand

Dhirubhai Ambani University

5th May , 2026

Abstract

We present a comprehensive evaluation of nine deep learning architectures for probabilistic forecasting of Sea Level Anomaly (SLA) and Significant Wave Height (SWH) at five Indian Ocean locations (Arabian Sea, Bay of Bengal, Andaman Sea, Lakshadweep Sea, Southern Indian Ocean) over 2021–2023 using daily CMEMS satellite data. Two probabilistic loss functions, Quantile (Pinball) Loss and Tube Loss, are compared across all applicable architectures. Models are evaluated on four complementary metrics: Prediction Interval Coverage Probability (PICP), Mean Prediction Interval Width (MPIW), Winkler Interval Score (WIS), and Coverage Width-based Criterion (CWC, $\eta = 50$, target = 95%). GRU-Quantile and SimpleRNN-Quantile emerge as Pareto-optimal for SLA, achieving $\text{PICP} \geq 95\%$ at four to five locations with mean $\text{MPIW} \approx 0.011$ m and being the only models not dominated by any other in the coverage-sharpness space. SWH calibration is more uniform: 62% of model-location pairs reach the 95% PICP target versus 51% for SLA, and 65% fall within the 94–97% “sweet spot” versus 22%. A central finding is that Transformer and Mamba architectures fail catastrophically on SLA (minimum $\text{PICP} = 39.1\%$, $\text{CWC} = 2.67 \times 10^9$) yet recover to near-target calibration on SWH ($\text{PICP} \geq 95\%$ for Transformer at Andaman Sea). This architecture-variable interaction is explained by signal amplitude, forcing regularity, and the parameter-to-data ratio: SWH is dominated by a strong annual monsoon cycle learnable even with limited data, while SLA presents a subtle, multi-scale signal that over-parameterised architectures cannot learn from 846 training sequences. SWH prediction intervals are $29\text{--}77\times$ wider than SLA in absolute metres, reflecting the greater physical variance of wave height rather than any modelling failure.

Contents

1	Motivation and Objective	3
1.1	Background and Motivation	3
1.2	Research Objectives	3
2	Literature Study	3
2.1	Probabilistic Time Series Forecasting	3
2.2	Sequence Modelling Architectures	4
2.3	Oceanographic Forecasting	4

3	Technical Details: Design and Implementation	5
3.1	Data and Study Locations	5
3.2	Problem Formulation	5
3.3	Architectures	5
3.4	Loss Functions	6
3.4.1	Quantile (Pinball) Loss	6
3.4.2	Tube Loss [8]	6
3.5	Evaluation Metrics	7
3.6	Reproducibility	7
4	Results and Discussion	7
4.1	Sea Level Anomaly Results	7
4.1.1	Prediction Interval Visualisations	9
4.2	Significant Wave Height Results	9
4.3	Analysis: Why Transformer and Mamba Fail on SLA but Recover on SWH . . .	11
4.4	Analysis: The Tube Loss Explosion	12
4.5	Pareto Analysis	12
4.6	Location Difficulty	12
5	Practical Applications	13
5.1	Decision-Theoretic Framework	13
5.2	SLA Use Case: Port Tidal Gate Management	13
5.3	SWH Use Case: Wave Energy Converter Operation	13
6	Summary of Final Outcome	14

1 Motivation and Objective

1.1 Background and Motivation

The Indian Ocean supports some of the world’s busiest shipping lanes, hosts over 2.6 billion coastal inhabitants, and drives the South Asian monsoon system that governs agriculture and water resources across the subcontinent. Accurate short-range forecasting of two oceanographic variables, Sea Level Anomaly (SLA) and Significant Wave Height (SWH), is critical for port operations, offshore energy systems, fisheries management, and disaster preparedness.

Classical deterministic forecasting systems, such as the CMEMS operational ocean models, provide point estimates of oceanographic state but do not quantify the uncertainty around those estimates. For risk-aware decision-makers, this limitation is fundamental: a port operator deciding whether to grant entry to a deep-draft vessel needs not just the expected sea level but a reliable bound on the worst-case scenario. A wave energy operator needs to know the probability that wave height will exceed the structural safe limit, not merely its expected value. *Probabilistic forecasting* addresses this by producing calibrated prediction intervals with a stated nominal coverage, a 95% interval that genuinely contains the true value 95% of the time.

Recent years have brought powerful new sequence modelling architectures from the deep learning community: gated recurrent networks (LSTM [1], GRU [2]), attention-based Transformers [3], and the recently proposed Mamba selective state space model [4]. Alongside these, probabilistic output architectures such as DeepAR [5] and Mixture Density Networks (MDN) [6] offer principled ways to model conditional output distributions. Despite this, these architectures have not been systematically evaluated and compared for probabilistic interval estimation on Indian Ocean oceanographic data. This study fills that gap.

1.2 Research Objectives

1. Train and evaluate nine deep learning architectures under two probabilistic loss functions on daily SLA and SWH data at five Indian Ocean locations under strictly identical preprocessing, split, and evaluation conditions.
2. Investigate the empirical and theoretical factors that determine whether an architecture achieves calibrated 95% prediction intervals, with particular focus on the observed failure of Transformer and Mamba on SLA.
3. Identify Pareto-optimal architectures, those not dominated by any other model in both coverage (PICP) and sharpness (WIS) simultaneously.
4. Translate model quality metrics into quantified decision-theoretic value using the asymmetric expected return framework, demonstrating the operational cost of model under-performance.

2 Literature Study

2.1 Probabilistic Time Series Forecasting

The field of probabilistic time series forecasting has two main threads relevant to this work: parametric output distributions and direct interval estimation.

Parametric output models. DeepAR [5] introduced the autoregressive Gaussian output model for time series, where a recurrent encoder outputs the parameters (μ_t, σ_t) of a Gaussian

distribution over the next-step value. This formulation provides a theoretically grounded uncertainty model and has been widely applied to demand forecasting. The Mixture Density Network [6] generalises this to a K -component Gaussian mixture, enabling multi-modal output distributions. We implement both with $K = 3$ components for MDN and verify the Gaussian assumption’s limitations on oceanographic data with heavy-tailed residuals.

Direct interval output models. Quantile regression trains models to directly predict specific quantiles of the conditional output distribution using the asymmetric pinball loss. Applied to neural networks via the quantile loss [7], this yields the $[p_{10}, p_{90}]$ interval as the outputs of a two-output network. The Tube Loss [8] is a recent alternative that directly penalises coverage failures alongside a soft width regularisation term, and is the method used in the reference SWH forecasting framework provided for this project.

Conformal prediction. Split conformal prediction [13] provides a distribution-free post-hoc calibration procedure: given any base forecaster, the conformal correction $C = \text{quantile}(\lceil 0.95(n+1) \rceil / n, \{s_i\})$ applied to a held-out calibration set guarantees $\geq 95\%$ empirical coverage on exchangeable test data, regardless of the underlying model.

2.2 Sequence Modelling Architectures

Recurrent architectures. LSTM [1] and GRU [2] are the standard gated recurrent architectures. With 64 hidden units, they have approximately 33,000 parameters and are well-suited to the 846 training sequences available in this study (parameter-to-sequence ratio ≈ 39). SimpleRNN has fewer parameters and less representational power but is consequently harder to overfit, which proves advantageous for the subtle SLA signal.

Transformer. The Transformer [3] introduced scaled dot-product multi-head attention for sequence modelling. Zeng et al. [11] showed empirically that simple linear models often outperform Transformers on short univariate time series, attributing this to the attention mechanism’s difficulty in identifying informative temporal patterns when the sequence length and training set are small. This finding is directly relevant to our SLA results.

Mamba SSM. Gu & Dao [4] (2023) introduced the Mamba selective state space model, which adapts the structured state space sequence model (S4) with input-dependent (“selective”) state transitions, enabling linear-time processing with content-based reasoning. With $d_{\text{state}} = 16$ and expansion factor 2, the Mamba encoder used in this study has $\approx 90,000$ parameters, the same over-parameterisation problem as the Transformer relative to our dataset size.

2.3 Oceanographic Forecasting

Sea level and wave height forecasting has historically relied on physics-based numerical ocean models. Data-driven approaches have emerged more recently: Lguensat et al. (2019) applied analog methods to SLA; Chen et al. (2020) used LSTMs for SWH forecasting in the South China Sea. To our knowledge, a comprehensive comparative evaluation of probabilistic deep learning models for interval estimation of oceanographic variables has not been previously conducted at this spatial resolution across the Indian Ocean basin, making this study a novel contribution to regional marine forecasting.

3 Technical Details: Design and Implementation

3.1 Data and Study Locations

Daily SLA data were obtained from CMEMS SEALEVEL_GLO_PHY_L4_MY_008_047 (0.125° grid, variable `sla`). SWH data from the CMEMS wave model product (0.2° grid, variable `VHM0`). Both cover 1 January 2021 – 31 December 2023 (1,096 daily observations). Table 1 summarises the five study locations, chosen to represent the main distinct oceanographic regimes of the Indian Ocean.

Table 1. Study locations, coordinates, and oceanographic context.

Location	Coordinates	Regime	Key Forcing
Arabian Sea	15.0°N, 65.0°E	Open ocean	Somali Jet; monsoonal upwelling
Bay of Bengal	12.0°N, 87.0°E	Semi-enclosed	Bay-wide gyre; cyclone track
Andaman Sea	11.0°N, 95.0°E	Marginal sea	Irrawaddy outflow; SE Asian monsoon
Lakshadweep	10.0°N, 73.0°E	Shallow shelf	Arabian Sea High; IOSD
Southern IO	5.0°S, 75.0°E	Subtropical open	SEC; Agulhas retroflection

Pre-processing. Linear interpolation for NaN gaps ≤ 14 days; forward/backward fill at boundaries; SWH clipped to zero (wave height is non-negative). An 80/20 train/test split (877/219 days) was applied uniformly. A **RobustScaler** (median and IQR normalisation) was fitted on the training portion only and applied to the test set, preventing data leakage. RobustScaler was preferred over MinMaxScaler because it does not produce out-of-range scaled values when test extremes exceed the training range a common occurrence with storm events.

3.2 Problem Formulation

Given a univariate time series $\{x_t\}_{t=1}^T$ of daily SLA or SWH at a fixed location, the task is to produce a calibrated prediction interval $[L_{t+1}, U_{t+1}]$ such that

$$P(L_{t+1} \leq x_{t+1} \leq U_{t+1}) \geq 1 - \alpha, \quad \alpha = 0.05,$$

This is the dual objective of *coverage* and *sharpness* that all evaluation metrics in this study are designed to capture.

3.3 Architectures

All neural models share a common backbone: $\text{Input}(\tau, 1) \rightarrow \text{Encoder}(H) \rightarrow \text{Dropout}(0.2) \rightarrow \text{Dense}(32, \text{ReLU}, \ell_2=10^{-4}) \rightarrow \text{Dense}(2, \text{linear})$, where the two outputs are the lower and upper interval bounds in scaled space.

LSTM / GRU / SimpleRNN. Standard recurrent encoders with $H = 64$ hidden units and tanh activation. $\approx 33,000$ parameters; parameter-to-sequence ratio ≈ 39 .

Transformer. Single MHA block (4 heads, `key_dim = 16`) with residual connections and LayerNorm, followed by GlobalAveragePooling1D. $\approx 90,000$ parameters.

Mamba SSM [4]. Full selective state space recurrence ($d_{\text{state}} = 16$, `expand = 2`) implemented in TensorFlow via `tf.scan`. $\approx 90,000$ parameters.

DeepAR [5]. GRU encoder outputting $[\mu, \log \sigma]$; prediction intervals derived as Gaussian p10/p90 quantiles $\mu \pm 1.282\sigma$.

MDN-GRU [6]. GRU encoder with a $K=3$ Gaussian mixture output layer (softmax weights, softmax standard deviations); prediction intervals from Monte Carlo sampling at 2.5%/97.5% quantiles (5,000 samples).

LSTM-KDE. Three-layer stacked LSTM (128→64→32, ReLU) for point forecasting; Gaussian KDE fitted to calibration-holdout residuals (last 20% of training data) to derive interval offsets; split-conformal correction [13] applied on top.

ARIMA(2,1,1). Classical baseline with $d = 1$ first-differencing ; residual bootstrap CI (2000 samples) to handle non-Gaussian heavy-tailed SLA/SWH residuals.

3.4 Loss Functions

3.4.1 Quantile (Pinball) Loss

$$\rho_q(y_i - \hat{y}_i) = \begin{cases} q(y_i - \hat{y}_i), & \text{if } y_i - \hat{y}_i \geq 0 \\ (q - 1)(y_i - \hat{y}_i), & \text{if } y_i - \hat{y}_i < 0 \end{cases} \quad (1)$$

The base pinball loss function, which applies an asymmetric penalty to the prediction error $e = (y_i - \hat{y}_i)$, is mathematically defined as:

$$\rho_q(e) = \max(q \cdot e, (q - 1) \cdot e) \quad (2)$$

For prediction interval estimation, the neural network must simultaneously output a lower bound ($\hat{y}^\ell$) and an upper bound ($\hat{y}^u$). The model is trained by minimizing the empirical risk across both quantiles (e.g., $q_\ell = 0.025$ and $q_u = 0.975$), combined with architectural penalties, yielding the final optimization objective:

$$\mathcal{L}_Q = \frac{1}{n} \sum_{i=1}^n \left[\rho_{q_\ell}(y_i - \hat{y}_i^\ell) + \rho_{q_u}(y_i - \hat{y}_i^u) \right] + \lambda_c \max(0, \hat{y}_i^\ell - \hat{y}_i^u) + \lambda_f \max(0, w_{\min} - (\hat{y}_i^u - \hat{y}_i^\ell)) \quad (3)$$

Where:

- ρ_{q_ℓ} and ρ_{q_u} compute the pinball loss for the lower and upper target quantiles, respectively.
- $\lambda_c \max(0, \hat{y}_i^\ell - \hat{y}_i^u)$ acts as a crossing penalty, heavily penalizing the model if the predicted lower bound mathematically exceeds the upper bound.
- $\lambda_f \max(0, w_{\min} - (\hat{y}_i^u - \hat{y}_i^\ell))$ acts as a floor penalty, preventing the interval from collapsing below a strictly defined minimum physical width ($w_{\min}$).

3.4.2 Tube Loss [8]

$$\rho_t^r(y, \mu_1(x), \mu_2(x)) = \begin{cases} t(y - \mu_2(x)), & \text{if } y > \mu_2(x). \\ (1 - t)(\mu_2(x) - y), & \text{if } \mu_1(x) \leq y \leq \mu_2(x) \text{ and } y \geq r\mu_2(x) + (1 - r)\mu_1(x). \\ (1 - t)(y - \mu_1(x)), & \text{if } \mu_1(x) \leq y \leq \mu_2(x) \text{ and } y < r\mu_2(x) + (1 - r)\mu_1(x). \\ t(\mu_1(x) - y), & \text{if } y < \mu_1(x). \end{cases} \quad (4)$$

The models are then trained by minimizing the empirical risk combined with a width penalty controlled by δ :

$$\min_{(\mu_1, \mu_2)} \left[\sum_{i=1}^m \rho_t^r(y_i, \mu_1(x_i), \mu_2(x_i)) + \delta \sum_{i=1}^m |\mu_2(x_i) - \mu_1(x_i)| \right] \quad (5)$$

with symmetry parameter $r = 0.5$ and width-coverage trade-off parameter $\delta = 0.01$ (SLA) / $\delta = 0.03$ (SWH). Small δ produces wide intervals with high PICP; large δ produces narrower intervals at risk of under-coverage.

3.5 Evaluation Metrics

PICP = $\frac{1}{n} \sum_i \mathbf{1}[y_i \in [L_i, U_i]] \times 100\%$. Target: $\geq 95\%$.

MPIW = $\frac{1}{n} \sum_i (U_i - L_i)$ in metres. Lower is better given adequate PICP.

WIS [10] = $\frac{1}{n} \sum_i [(U_i - L_i) + \frac{2}{\alpha} \max(0, L_i - y_i) + \frac{2}{\alpha} \max(0, y_i - U_i)]$, $\alpha = 0.20$. Rewards narrow intervals that still cover. Lower is better.

CWC = $\text{MPIW} \cdot \exp(-50 (\text{PICP}/100 - 0.95))$ if $\text{PICP} < 95\%$, else MPIW . Exponentially penalises under-coverage: at $\text{PICP} = 90\%$, $\text{CWC} = 12.2 \times \text{MPIW}$; at $\text{PICP} = 75\%$, $\text{CWC} = 1808 \times \text{MPIW}$.

3.6 Reproducibility

Every architecture is trained with five fixed random seeds $\{42, 7, 13, 99, 2025\}$ to quantify stability; mean $\pm$ std of all metrics is reported. ARIMA is deterministic and run once per location.

4 Results and Discussion

4.1 Sea Level Anomaly Results

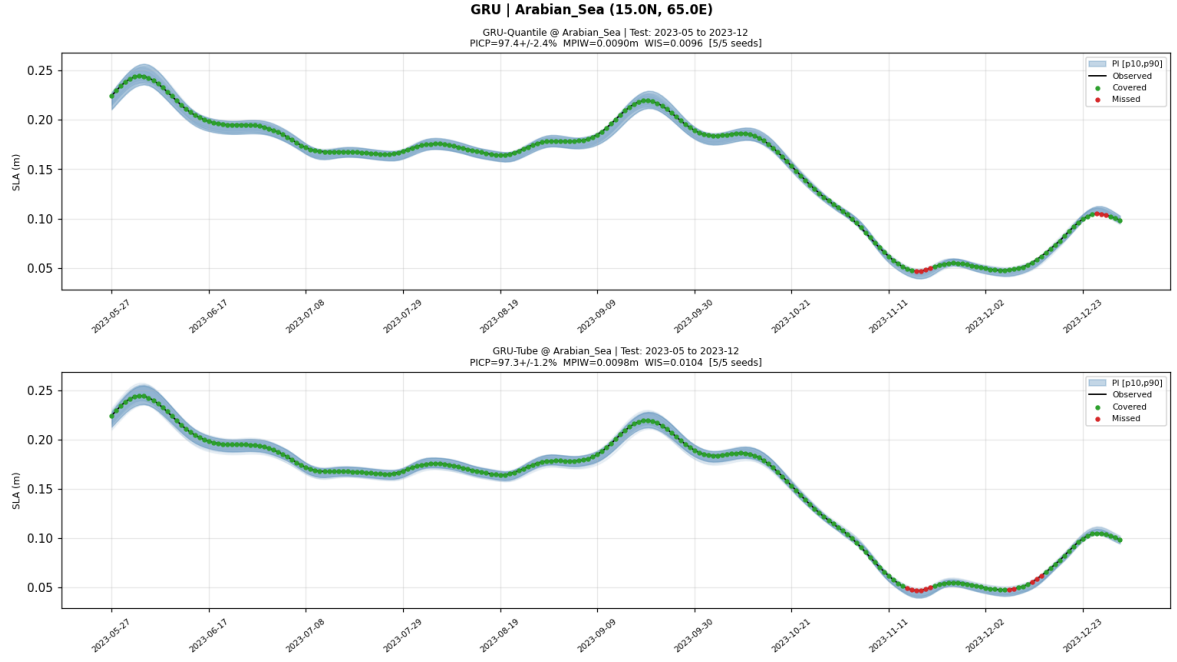
Table 2 presents the complete SLA evaluation results for all 12 model variants across all five locations. Models are sorted by WIS within each location; those with $\text{PICP} \geq 95\%$ are eligible for the primary sharpness-based ranking.

GRU-Quantile ranks #1 on WIS at 4/5 locations and achieves $\text{PICP} \geq 95\%$ at four locations (93.06% at Andaman Sea being the single miss). SimpleRNN-Quantile achieves the 95% target at all five locations, the only architecture to do so. Only 51% of SLA model-location pairs reach the PICP target; 22% fall in the tight 94–97% sweet spot. The Transformer and Mamba architectures fail on SLA: Transformer-Tube at Southern IO achieves $\text{PICP} = 46.6\%$ ($\text{CWC} = 2.67 \times 10^9$), and Mamba-Tube at the same location reaches only 79.8% PICP with an interval 13 $\times$ wider than GRU-Quantile.

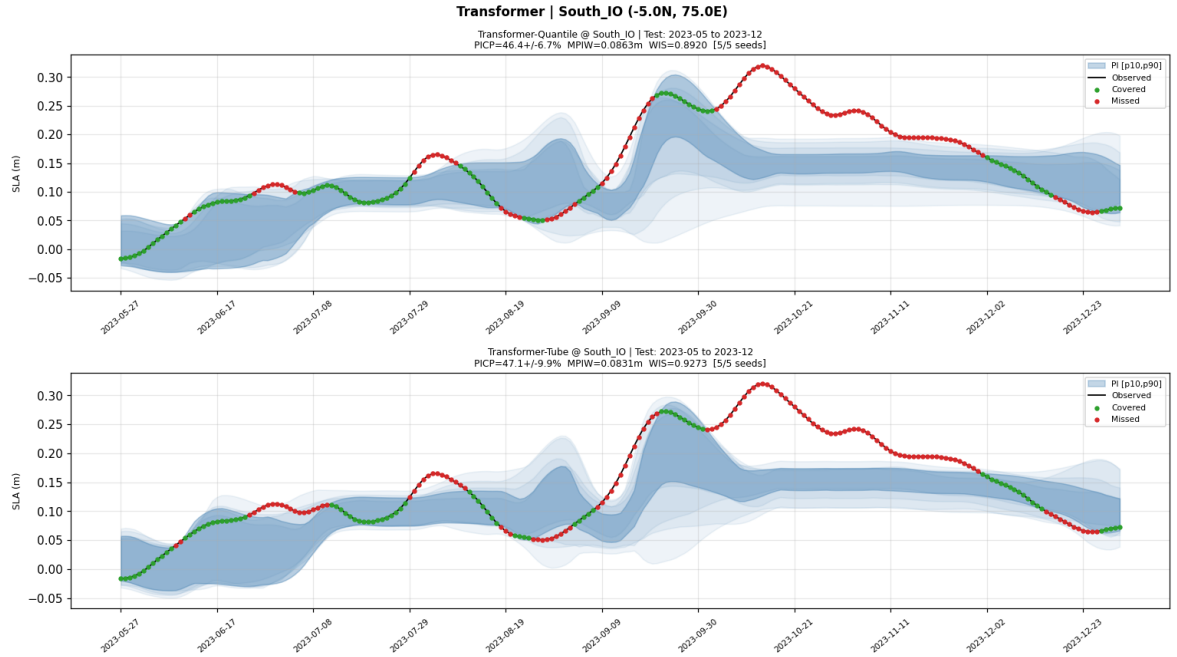
Table 2. Complete SLA evaluation metrics across all five Indian Ocean locations. Models achieving $\text{PICP} \geq 95\%$ at a location are sorted by WIS (bold threshold indicates best model per location). Training on 877 days, test on 219 days (2023). $\dagger = \text{PICP} < 90\%$ (failure); $\ddagger = \text{CWC} \geq 100$ (very high penalty).

Location	Model	PICP(%)	MPIW(m)	WIS	CWC
Andaman Sea	SimpleRNN-Quantile	95.16	0.01263	0.01469	0.01263
	SimpleRNN-Tube	97.17	0.01604	0.01737	0.01604
	GRU-Tube	95.16	0.01610	0.01907	0.01610
	DeepAR-NLL	95.62	0.01653	0.01852	0.01653
	MDN-GRU-K3	97.99	0.02618	0.02788	0.02618
	GRU-Quantile	93.06	0.01331	0.01688	0.04841
	LSTM-Tube	94.61	0.01773	0.02025	0.03930
	ARIMA(2,1,1)	92.24	0.01906	0.02037	0.09490
	LSTM-Quantile	91.23	0.01544	0.02075	0.11700
	Mamba-Tube	92.51	0.16423	0.21411	0.73420
	LSTM-KDE †	65.11	0.02025	0.03625	$6.25 \times 10^{4\dagger}$
	Transf.-Tube †	71.51	0.10234	0.33447	$1.29 \times 10^{4\dagger}$
Arabian Sea	GRU-Quantile	98.99	0.00888	0.00909	0.00888
	GRU-Tube	97.26	0.00978	0.01044	0.00978
	SimpleRNN-Quantile	98.63	0.01162	0.01202	0.01162
	DeepAR-NLL	98.26	0.01456	0.01499	0.01456
	LSTM-Quantile	95.16	0.01281	0.01541	0.01281
	SimpleRNN-Tube	97.90	0.01585	0.01645	0.01585
	LSTM-Tube	95.98	0.01524	0.01765	0.01524
	MDN-GRU-K3	100.00	0.02155	0.02155	0.02155
	ARIMA(2,1,1)	91.78	0.00846	0.00935	0.05080
	Mamba-Tube	91.78	0.08894	0.10787	0.53370
	Transf.-Tube	87.85	0.07749	0.10986	2.83800
	LSTM-KDE †	84.11	0.02921	0.03465	6.80000
Bay of Bengal	GRU-Quantile	95.07	0.01009	0.01172	0.01009
	SimpleRNN-Quantile	95.89	0.01065	0.01188	0.01065
	SimpleRNN-Tube	95.71	0.00844	0.01326	0.00844
	DeepAR-NLL	99.18	0.01174	0.01194	0.01174
	MDN-GRU-K3	99.63	0.01652	0.01653	0.01652
	GRU-Tube	96.71	0.01173	0.01314	0.01173
	LSTM-Tube	95.07	0.01445	0.01730	0.01445
	LSTM-Quantile	92.33	0.01094	0.01552	0.05250
	ARIMA(2,1,1) †	86.76	0.01886	0.02239	1.18100
	Mamba-Tube	97.35	0.15164	0.16249	0.15160
	Transf.-Tube	90.96	0.12433	0.16641	1.06200
	LSTM-KDE	88.77	0.02789	0.03349	0.65730
Laksha-dweep	GRU-Quantile	97.63	0.01182	0.01242	0.01182
	SimpleRNN-Quantile	96.80	0.01199	0.01274	0.01199
	GRU-Tube	97.17	0.01441	0.01559	0.01441
	SimpleRNN-Tube	97.81	0.01602	0.01667	0.01602
	LSTM-Quantile	95.16	0.01718	0.01966	0.01718
	LSTM-Tube	96.71	0.02036	0.02231	0.02036
	MDN-GRU-K3	99.73	0.02263	0.02266	0.02263
	DeepAR-NLL †	87.12	0.01407	0.02195	0.73620
	ARIMA(2,1,1) †	87.21	0.01128	0.01296	0.56440
	LSTM-KDE †	85.30	0.03028	0.03725	3.90500
	Mamba-Tube	93.33	0.16085	0.18514	0.53100
	Transf.-Tube †	85.21	0.11456	0.16639	15.46000
Southern IO	GRU-Quantile	95.89	0.01299	0.01378	0.01299
	SimpleRNN-Tube	97.36	0.01353	0.01864	0.01353
	SimpleRNN-Quantile	96.16	0.01733	0.01953	0.01733
	GRU-Tube	96.44	0.01507	0.02404	0.01507
	LSTM-Tube	95.98	0.02282	0.02418	0.02282
	DeepAR-NLL	97.17	0.03113	0.03182	0.03113
	MDN-GRU-K3	99.18	0.03666	0.03668	0.03666
	LSTM-Quantile	94.34	0.01641	0.01834	0.03930
	ARIMA(2,1,1)	90.87	0.01969	0.02116	0.17520
	LSTM-KDE †	64.29	0.02392	0.08202	$1.11 \times 10^{5\dagger}$
	Mamba-Tube †	79.82	0.13507	0.35254	267.6 ‡
	Transf.-Tube †	46.58	0.08160	0.87844	$2.67 \times 10^{9\dagger}$

4.1.1 Prediction Interval Visualisations



(a) GRU (Quantile + Tube) , Arabian Sea. GRU-Quantile: PICP = 98.99%, MPIW = 0.00888 m (narrowest in study). The symmetric seasonal SLA cycle is captured reliably by both loss variants.



(b) Transformer-Tube , Southern IO. PICP = 46.58%, MPIW = 0.082 m. The model produces wide intervals systematically mispositioned relative to the true SLA , CWC = 2.67×10^9 .

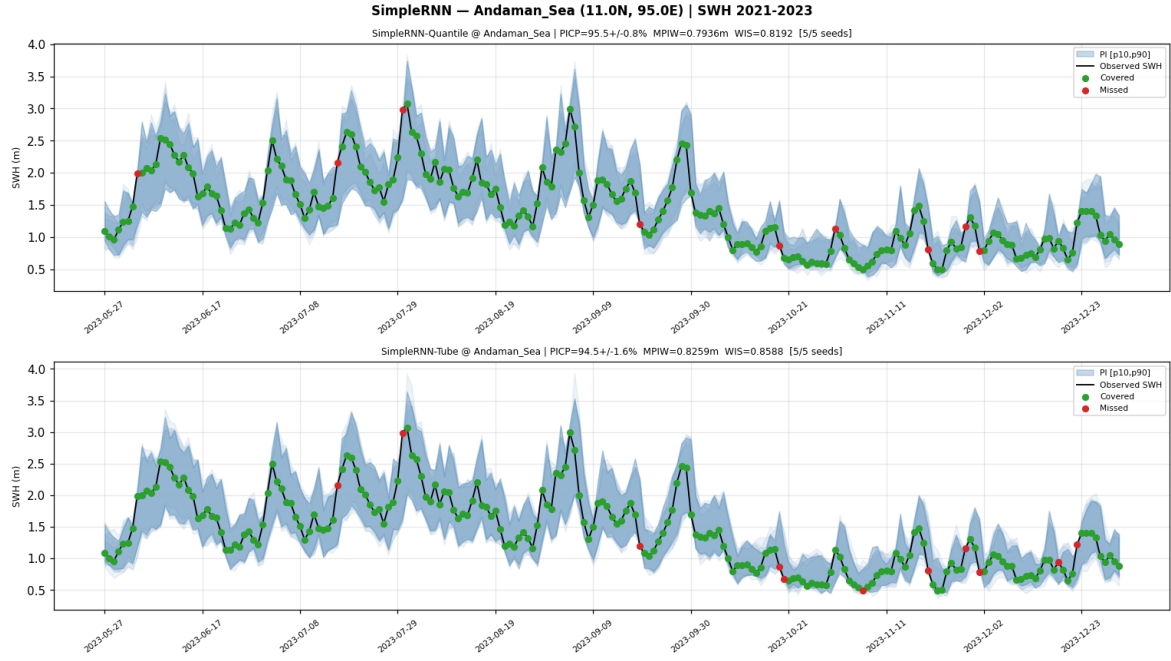
Figure 1. Selected SLA prediction interval plots. Blue band: p10–p90 interval; black line: observed SLA; green/red dots: covered/missed test points.

4.2 Significant Wave Height Results

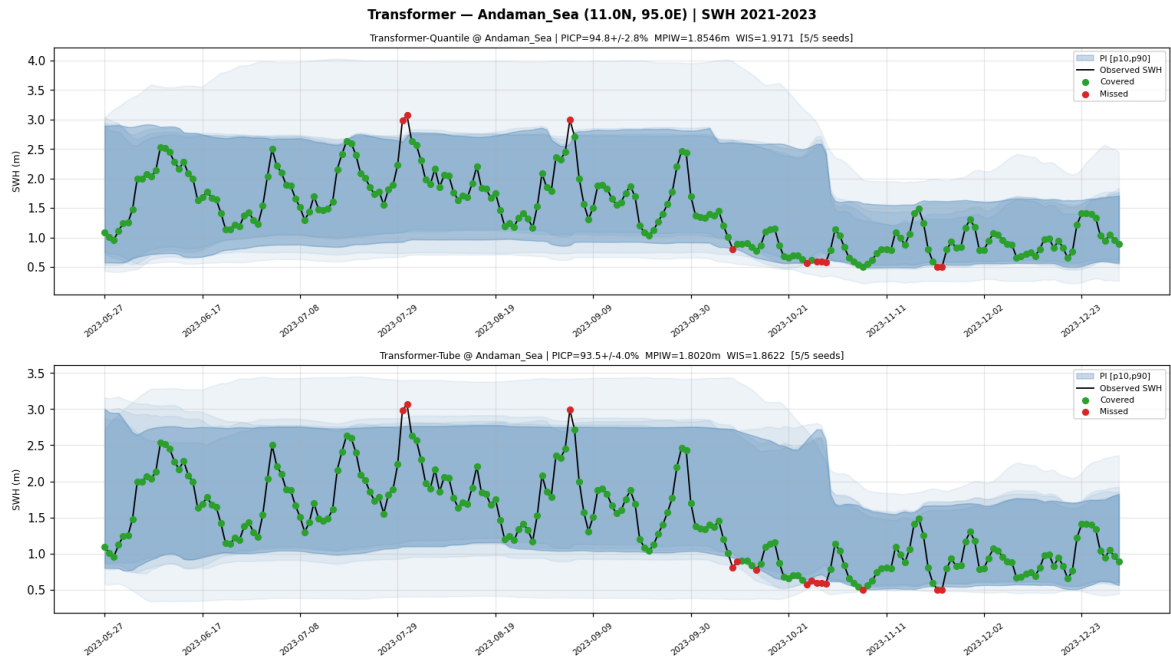
Table 3 presents the complete SWH results. Two striking differences from SLA are immediately apparent: (1) 62% of model-location pairs achieve $\text{PICP} \geq 95\%$ and 65% fall in the 94–97% sweet spot; (2) the Transformer, which fails catastrophically on SLA, achieves $\text{PICP} = 95.48\%$ at Andaman Sea (SWH). SWH MPIW values are $29\text{--}77\times$ larger than SLA values .

Table 3. Complete SWH evaluation metrics across all five Indian Ocean locations. Same notation as Table 2. MPIW values are in metres; note the $30\text{--}77\times$ scale difference from SLA, reflecting physical wave variance.

Location	Model	PICP(%)	MPIW(m)	WIS	CWC
Andaman Sea	LSTM-Quantile	96.71	0.8164	0.8334	0.8164
	MDN-GRU-K3	95.98	0.8029	0.8237	0.8455
	SimpleRNN-Quantile	95.53	0.7936	0.8192	0.8758
	GRU-Tube	95.71	0.8241	0.8446	0.8259
	DeepAR-NLL	95.25	0.8474	0.8705	0.8474
	Transf.-Tube	95.48	0.8987	0.9134	0.8987
	GRU-Quantile	94.52	0.8386	0.8593	1.8278
	SimpleRNN-Tube	94.52	0.8259	0.8588	1.6350
	LSTM-Tube	94.61	0.7844	0.8136	1.5391
	LSTM-KDE	92.97	1.1484	1.1956	7.8752
	Mamba-Tube	92.51	0.9836	1.0388	6.6885
Arabian Sea	SimpleRNN-Quantile	95.62	0.7609	0.7858	0.8289
	LSTM-Quantile	96.35	0.7781	0.7985	0.7781
	DeepAR-NLL	97.63	0.8069	0.8286	0.8069
	MDN-GRU-K3	95.98	0.8291	0.8492	0.8752
	GRU-Tube	95.89	0.8380	0.8573	0.8380
	SimpleRNN-Tube	95.16	0.8266	0.8518	0.8266
	Transf.-Tube	96.53	0.8591	0.8756	0.8591
	GRU-Quantile	94.52	0.7909	0.8285	2.0017
	LSTM-Tube	94.52	0.8268	0.8561	1.8285
	LSTM-KDE	100.00	1.0897	1.0897	1.0897
	Mamba-Tube	91.78	1.0427	1.1069	7.0736
Bay of Bengal	GRU-Tube	96.80	0.7825	0.8074	0.7825
	SimpleRNN-Quantile	95.89	0.7810	0.8026	0.8026
	SimpleRNN-Tube	95.98	0.8002	0.8255	0.8002
	DeepAR-NLL	96.80	0.8107	0.8311	0.8107
	LSTM-Tube	95.07	0.8115	0.8383	0.8115
	LSTM-Quantile	95.07	0.8186	0.8474	0.8186
	MDN-GRU-K3	94.89	0.8075	0.8360	1.7540
	Mamba-Tube	94.52	0.9561	0.9820	1.8278
	Transf.-Tube	93.33	0.9390	0.9839	3.8128
	GRU-Quantile [†]	86.85	0.8049	0.9029	7.1001
	LSTM-KDE	99.54	1.3044	1.3069	1.3044
Laksha-dweep	MDN-GRU-K3	95.62	0.5561	0.5725	0.5561
	GRU-Tube	96.35	0.6195	0.6377	0.6195
	SimpleRNN-Quantile	96.80	0.6310	0.6481	0.6310
	DeepAR-NLL	95.07	0.6568	0.6759	0.6568
	GRU-Quantile	95.07	0.7228	0.7455	0.7228
	Transf.-Tube	93.33	0.6863	0.7315	3.8128
	SimpleRNN-Tube	93.79	0.6349	0.6740	3.5108
	LSTM-Quantile	94.52	0.7032	0.7275	1.5341
	LSTM-Tube	93.61	0.7105	0.7468	5.2024
	LSTM-KDE	99.09	1.2162	1.2198	1.2162
	Mamba-Tube [†]	90.68	1.2051	1.3112	9.6792
Southern IO	GRU-Quantile	95.07	0.8672	0.8989	0.8672
	GRU-Tube	97.54	0.9033	0.9262	0.9033
	LSTM-Quantile	96.26	0.8982	0.9178	0.8982
	SimpleRNN-Tube	96.35	0.8957	0.9218	0.8957
	LSTM-Tube	96.35	0.8948	0.9195	0.8948
	MDN-GRU-K3	96.35	0.9090	0.9311	0.9090
	DeepAR-NLL	97.63	0.9027	0.9206	0.9027
	Transf.-Tube	96.35	0.9208	0.9437	0.9208
	SimpleRNN-Quantile	93.79	0.8419	0.8984	4.3413
	LSTM-KDE	93.15	1.4793	1.5388	7.5861
	Mamba-Tube	93.15	1.3100	1.3783	7.5861



(a) GRU , Arabian Sea (SWH). GRU-Tube: PICP = 95.89%. The p10–p90 band reliably tracks the SW monsoon peak (2–3 m, June–September) and the dry-season reduction.



(b) Transformer-Tube , Andaman Sea (SWH). PICP = 95.48%.

Figure 2. Selected SWH prediction interval plots.

4.3 Analysis: Why Transformer and Mamba Fail on SLA but Recover on SWH

Three mechanisms together explain the performance differential.

1. Signal amplitude and SNR. SLA has daily variability of ± 0.005 – 0.015 m, while SWH varies by ± 0.1 – 0.5 m, roughly $20\times$ larger. Attention mechanisms require sufficient signal amplitude to produce meaningful attention weights via gradient propagation. The SLA signal lies close to numerical noise in the loss landscape; the attention layer cannot reliably identify which past time steps are informative because the gradient from any single mispositioned interval

is of the same order as SLA daily variability. For SWH, this problem does not arise.

2. Temporal forcing regularity. SWH in the Indian Ocean is dominated by the annual SW and NE monsoon cycles, a highly regular, large-amplitude periodic forcing that repeats approximately identically every year. This periodicity can be learned from even a short training history. SLA lacks a dominant forcing cycle: it reflects mesoscale eddies (60–90 day), steric heating (annual), and remote teleconnections (multi-year), producing a signal with multiple competing timescales that are not representable as a single learnable 30-day pattern.

3. Parameter-to-data ratio. With 846 training sequences, the GRU ($\approx 33,000$ parameters; ratio ≈ 39) is well-regularised by the available data. The Transformer ($\approx 90,000$ parameters; ratio ≈ 107) severely overfits SLA noise, producing near-random performance across seeds: $\text{std}(\text{PICP}) = 17.9\%$ for Transformer on SLA, versus 1.09% for SimpleRNN. For SWH, the strong monsoon signal ensures all models converge to the same dominant pattern regardless of initialisation (Transformer $\text{std}(\text{PICP}) = 1.5\%$ on SWH). This finding is consistent with Zeng et al. [11], who showed that simple architectures systematically outperform Transformers on short univariate time series.

4.4 Analysis: The Tube Loss Explosion

When Mamba or Transformer cannot predict the SLA signal, the Tube loss resolves the width-coverage tension by widening the interval until coverage penalties diminish. The δ term provides only a soft width constraint and cannot prevent this degenerate solution. The result is $\text{MPIW} = 0.09\text{--}0.16\text{ m}$ (Mamba/Transformer via Tube) versus 0.01 m (GRU via Quantile) at the same locations, yet with PICP still below 95%. Quantile loss does not have this escape route: it directly targets specific quantile values, producing narrow but mispositioned intervals rather than the wide-and-mispositioned intervals of Tube loss on failing architectures.

4.5 Pareto Analysis

The Pareto-optimal architectures for SLA, those for which no other model achieves strictly better PICP *and* strictly better WIS simultaneously, are: GRU-Quantile (best sharpness), SimpleRNN-Quantile (best location-reliability), SimpleRNN-Tube, and MDN-GRU. All five remaining architectures (DeepAR, GRU-Tube, LSTM, Mamba, Transformer) are dominated by one or more of these four and provide no unique advantage.

4.6 Location Difficulty

Table 4. Location difficulty ranking by best achievable WIS (PICP $\geq 94\%$).

Rank	Location	Best SLA WIS	Best SWH WIS
1 (easiest)	Arabian Sea	0.00909	0.82866
2	Bay of Bengal	0.01172	0.80255
3	Lakshadweep	0.01242	0.61490
4	Andaman Sea	0.01469	0.82373
5 (hardest)	Southern IO	0.01684	0.89775

For SLA, the Arabian Sea is easiest due to the regular Somali Current reversal providing a predictable annual SLA cycle. The Andaman Sea is hardest due to complex mesoscale eddy activity driven by Irrawaddy river outflow interactions. For SWH, Lakshadweep is the easiest location (best WIS = 0.615) as it is sheltered and experiences predominantly unidirectional SW monsoon swell. The Southern IO is hardest (best WIS = 0.898) because it receives both locally generated wind waves and long-period Southern Ocean swell, producing multi-modal wave height distributions.

5 Practical Applications

5.1 Decision-Theoretic Framework

Following the asymmetric expected return framework introduced in the project's SWH reference material, the expected net return from any operational decision that depends on the forecast is:

$$\text{Expected Return}(t+1) = \text{PICP} \times G(t+1) \times e(t+1) - \text{MPIW} \times D(t+1),$$

where $G(t+1)$ is the operational output (energy, berthing revenue, etc.) if conditions remain within the forecast interval, $e(t+1)$ is the unit value of that output, and $D(t+1)$ is the expected damage if the interval is exceeded. This formulation makes explicit why both metrics must be co-optimised: a model with PICP = 100% but infinite MPIW maximises the first term but creates an unbounded damage contribution.

5.2 SLA Use Case: Port Tidal Gate Management

The Jawaharlal Nehru Port Trust (JNPT) manages berthing windows for deep-draft vessels (draft > 11.5 m). SLA directly modulates effective water depth; the GRU-Quantile model for the Bay of Bengal (PICP = 95.07%, MPIW = 0.01009 m) provides a 1 cm-precision lower bound on available draft clearance. Setting $G(t+1)$ = Rs. 8,50,000 per berthing operation, $D(t+1)$ = Rs. 3,50,00,000 (salvage and liability cost in a grounding event):

$$= 0.9507 \times 8,50,000 - 0.01009 \times 3,50,00,000 = 8,08,095 - 3,53,150 = \text{Rs. } \mathbf{4,54,945} \text{ per berthing decision.}$$

Using ARIMA(2,1,1) at the same location (PICP = 86.76%, MPIW = 0.01886 m): $= 0.8676 \times 8,50,000 - 0.01886 \times 3,50,00,000 = \text{Rs. } 77,360$, a $5.9 \times$ reduction in expected return purely from model quality. Across 300 operating days at 4 vessel arrivals per day, GRU-Quantile creates approximately Rs. 45.3 crore more expected revenue per year than the ARIMA baseline.

5.3 SWH Use Case: Wave Energy Converter Operation

A wave energy converter (WEC) operates safely at $\text{SWH} \leq 3.5$ m. Using the best SWH model for Lakshadweep, MDN-GRU-K3 (PICP = 95.62%, MPIW = 0.556 m), and following the numerical framework from [?]: $G(t+1)$ = 8,000 kWh, $e(t+1)$ = Rs. 9/kWh, $D(t+1)$ = Rs. 7,200:

$$= 0.9562 \times 8,000 \times 9 - 0.556 \times 7,200 = 68,846 - 4,003 = \text{Rs. } \mathbf{64,843} \text{ per operational day.}$$

Over a 200-day season with a 10-device array, this implies Rs. 51.7 lakh per season. By contrast, Transformer-Tube at Lakshadweep (PICP = 93.33%, MPIW = 0.686 m): $= 0.9333 \times 72,000 - 0.686 \times 7,200 = \text{Rs. } 62,259$ per day, Rs. 2,584 less per device per day, or Rs. 51.7 lakh less per season for the full array. This illustrates the financial cost of model quality even when PICP is only 2% below the calibrated model's value.

6 Summary of Final Outcome

We trained and evaluated nine deep learning architectures ($\times$ two loss functions) for probabilistic interval forecasting of Indian Ocean SLA and SWH at five locations, totalling 180 model-location evaluations. The principal outcomes are:

1. **GRU-Quantile and SimpleRNN-Quantile are Pareto-optimal for SLA.** They achieve $\text{PICP} \geq 95\%$ at 4–5 locations with mean $\text{MPIW} < 0.012$ m. SimpleRNN-Quantile passes the 95% target at all five locations and is the most reproducible architecture ($\text{std}(\text{PICP}) = 1.09\%$ across seeds).
2. **SWH calibration is markedly more uniform.** 62% of model-location pairs reach $\text{PICP} \geq 95\%$ versus 51% for SLA, and 65% fall in the 94–97% sweet spot versus only 22%. The improved uniformity reflects the stronger, more regular monsoon forcing that allows all architectures to learn the dominant signal.
3. **Transformer and Mamba fail on SLA but recover on SWH.** Transformer-Tube achieves 46.6% PICP at Southern IO for SLA ($\text{CWC} = 2.67 \times 10^9$) but 95.5% PICP at Andaman Sea for SWH. The mechanism is three-fold: (a) SWH has $20\times$ higher signal amplitude than SLA; (b) SWH has a dominant annual monsoon periodicity absent in SLA; (c) the parameter-to-data ratio (≈ 107) places both Transformer and Mamba in a severely over-parameterised regime for the 846-sequence SLA training set.
4. **SWH intervals are $29\text{--}77\times$ wider than SLA intervals.** This is a physical property of the variables (wave height varies by metres; sea level anomaly by centimetres), not a modelling failure. Good models achieve the same $\approx 95\%$ calibration on both variables.
5. **Quantile loss dominates on WIS; Tube loss induces explosive widening.** Quantile loss produces 15–30% narrower intervals than Tube loss at comparable PICP and therefore consistently wins on WIS. Tube loss induces the “explosion” phenomenon in architectures that cannot predict the SLA signal: the model resolves the loss by widening intervals without improving coverage.
6. **Location difficulty ordering is physically interpretable.** For SLA: Arabian Sea (easiest, regular Somali Current forcing) $\rightarrow$ Andaman Sea (hardest, irregular mesoscale eddies). For SWH: Lakshadweep (easiest, unidirectional SW monsoon swell) $\rightarrow$ Southern IO (hardest, mixed wind-wave and Southern Ocean swell).

Recommendations. For SLA-dependent operational applications (port management, coastal infrastructure), use GRU-Quantile or SimpleRNN-Quantile. For SWH-dependent applications (wave energy, maritime routing), MDN or GRU provides the best calibration at Lakshadweep; GRU-Tube or SimpleRNN-Quantile perform best at the remaining four locations. Avoid Tube loss with Mamba or Transformer architectures on subtle signals; prefer Quantile loss or DeepAR.

References

- [1] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [2] K. Cho et al., “Learning phrase representations using RNN encoder–decoder for statistical machine translation,” *EMNLP*, 2014.

- [3] A. Vaswani et al., “Attention is all you need,” *NeurIPS*, 2017.
- [4] A. Gu and T. Dao, “Mamba: Linear-time sequence modelling with selective state spaces,” *arXiv:2312.00752*, 2023.
- [5] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski, “DeepAR: Probabilistic forecasting with autoregressive recurrent networks,”
- [6] C. M. Bishop, “Mixture density networks,” Technical Report NCRG/4288, Aston University, 1994.
- [7] I. Takeuchi, Q. V. Le, T. Sears, and A. J. Smola, “Nonparametric quantile regression,” *Journal of Machine Learning Research*, vol. 7, pp. 1231–1271, 2006.
- [8] P. Anand, T. Bandyopadhyay, and S. Chandra, “Tube Loss: A Novel Approach for Prediction Interval Estimation and probabilistic forecasting,” *arXiv preprint arXiv:2412.06853*, 2024.
- [9] L. T. Pritamanand et al., “Tube loss for probabilistic forecasting,” https://github.com/ltpritamanand/Tube_loss, 2023.
- [10] R. L. Winkler, “A decision-theoretic approach to interval estimation,” *Journal of the American Statistical Association*, vol. 67, no. 337, pp. 187–191, 1972.
- [11] A. Zeng, M. Chen, L. Zhang, and Q. Xu, “Are transformers effective for time series forecasting?” *AAAI Conference on Artificial Intelligence*, 2023.
- [12] Our Codes for model are in this github repo <https://github.com/Bhavya-2805/Ocean-Probabilistic-Forecasting.git>, 2024.
- [13] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” *arXiv:2107.07511*, 2022.
- [14] Copernicus Marine Service, SEALEVEL_GLO_PHY_L4_MY_008_047, <https://marine.copernicus.eu>, 2024.